

HCSM-06

Sector Coarse Correspondence

The Gauge Algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ from the D_4 Endomorphism Algebra of the 14-Mode Multiplet The Hodge Complex Standard Model

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September 2026

Abstract

The Hodge Complex Standard Model assigns the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation. This paper derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a Lie subalgebra of the D_4 commutant on the complexified 14 mode multiplet, through the endomorphism algebra of the D_4 isotypic multiplicity spaces. We prove: (i) the D_4 commutant on $\mathbb{C}^{14}$ decomposes as $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$ with $\dim_{\mathbb{C}} = 27$; (ii) $\mathfrak{su}(3)$ embeds in the A_1 -block endomorphism algebra $M_3(\mathbb{C})$ as the traceless anti Hermitian subalgebra; (iii) $\mathfrak{su}(2)$ embeds in the B_1 -block endomorphism algebra $M_2(\mathbb{C})$; (iv) the five D_4 isotypic projectors span an abelian $\mathfrak{u}(1)^5$ subalgebra; (v) the generators of these three factors close to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, a 16-dimensional real Lie subalgebra of the commutant. On the full 56-dimensional Hodge sector at $\lambda = 4$, we derive the sector coarse correspondence: the space of D_4 -invariant, sector block diagonal, multiplicity scalar operators is exactly 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks. The identification of the A_1 factor with physical color and the B_1 factor with physical weak isospin is the framework's physical input, not a derivation. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The gauge algebra problem

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation [4]. On this multiplet, the D_4 point group of the square acts by permutation, and the paper HCSM-05 [5] established the D_4 isotypic decomposition.

The present paper asks: what Lie algebra emerges from the D_4 isotypic structure of this multiplet? The answer is the gauge algebra

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5, \quad (1)$$

realized as a Lie subalgebra of the D_4 commutant on the complexified 14-mode multiplet $\mathbb{C}^{14}$.

The derivation proceeds through the *sector-coarse correspondence*: the D_4 isotypic sectors of the mode multiplet correspond to the gauge-algebra factors through the endomorphism algebra of their multiplicity spaces. The “sector” structure is the fine D_4 isotypic decomposition; the “coarse” structure is the gauge algebra acting on the multiplicities. The correspondence between them is the content of this paper.

1.2 The one-line summary

The 14-mode multiplet at $\lambda = 4$ decomposes under D_4 as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$. The D_4 commutant on this multiplet is the endomorphism algebra $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$, of complex dimension 27. The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ embeds in the commutant: $\mathfrak{su}(3)$ acts on the A_1 multiplicity space, $\mathfrak{su}(2)$ on the B_1 multiplicity space, and the five isotypic projectors span the $\mathfrak{u}(1)^5$ center.

1.3 Main results

- (1) **D_4 isotypic decomposition** (Theorem 3.1, restated from HCSM-05 [5]). The 14-mode multiplet decomposes as

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (2)$$

of complex dimension 14.

- (2) **Commutant dimension** (Theorem 4.3). By Schur’s lemma, the D_4 commutant on V_{14} has complex dimension

$$\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = \sum_{\rho} m_{\rho}^2 = 3^2 + 1^2 + 2^2 + 2^2 + 3^2 = 27. \quad (3)$$

- (3) **Endomorphism algebra** (Theorem 4.4). The commutant is the endomorphism algebra

$$\text{Comm}(D_4, V_{14}) \cong \bigoplus_{\rho \in \{A_1, A_2, B_1, B_2, E\}} M_{m_{\rho}}(\mathbb{C}), \quad (4)$$

of complex dimension 27.

- (4) **$\mathfrak{su}(3)$ subalgebra** (Theorem 5.3). The traceless anti-Hermitian subalgebra of the A_1 -block factor $M_{m_{A_1}}(\mathbb{C}) = M_3(\mathbb{C})$ is a copy of $\mathfrak{su}(3)$, embedded in the commutant.
- (5) **$\mathfrak{su}(2)$ subalgebra** (Theorem 5.6). The traceless anti-Hermitian subalgebra of the B_1 -block factor $M_{m_{B_1}}(\mathbb{C}) = M_2(\mathbb{C})$ is a copy of $\mathfrak{su}(2)$, embedded in the commutant.
- (6) **$\mathfrak{u}(1)^5$ subalgebra** (Theorem 5.9). The five D_4 isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$ span an abelian Lie subalgebra of the commutant isomorphic to $\mathfrak{u}(1)^5$.
- (7) **Closure** (Theorem 5.10). The generators of the three factors above close to a Lie subalgebra of $\text{Comm}(D_4, V_{14})$ isomorphic to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, of real dimension $8 + 3 + 5 = 16$.
- (8) **Sector-coarse correspondence** (Theorem 6.3). On the full 56-dimensional Hodge sector at $\lambda = 4$, the space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators is exactly 15-dimensional, spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSMstack, whose numbering follows the master index in HCSM-00 [1]:

- HCSM-00 [1]: foundations, notation, and mapping.
- HCSM-01 [2]: the 8×8 torus substrate.
- HCSM-02 [3]: the Hodge complex on Λ .
- HCSM-04 [4]: the 14-mode multiplet.
- HCSM-05 [5]: orbit decomposition and inverse participation.
- HCSM-07 [6]: hypercharge quantization.

1.5 Organization

Section 2 states the substrate primitives. Section 3 states the D_4 isotypic decomposition (citing HCSM-05). Section 4 derives the commutant and its endomorphism structure. Section 5 derives the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$. Section 6 derives the sector-coarse correspondence on the 56-dimensional Hodge sector. Section 7 states what is derived and what is input. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup

2.1 The substrate and the mode multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (5)$$

At the self-paired eigenvalue $\lambda = 4$, the mode set M has 14 elements, listed explicitly in HCSM-04 [4]:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (6)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (7)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \quad (8)$$

We work with the complexified mode space $V_{14} = \mathbb{C}^{14}$, with basis $\{e_m : m \in M\}$. The physical mode vectors on the torus are real, but the complexification is the natural home of the gauge algebra (as in QFT).

2.2 The D_4 action

The D_4 point group of the square acts on Λ by

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{8}, \quad s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{8}, \quad (9)$$

subject to $r^4 = s^2 = (rs)^2 = e$. The permutation action of D_4 on the 14 modes extends to a representation of D_4 on V_{14} :

$$\rho_{14} : D_4 \rightarrow GL(V_{14}), \quad \rho_{14}(g)e_m = e_{g \cdot m}. \quad (10)$$

We denote by $R = \rho_{14}(r)$ and $S = \rho_{14}(s)$ the two generators.

2.3 The D_4 character table

The D_4 irreps are five: two 1-dimensional real representations A_1, A_2 , two 1-dimensional real representations B_1, B_2 , and one 2-dimensional real representation E . We use the standard conjugacy classes $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$ with sizes 1, 1, 2, 2, 2:

irrep ρ	e	r^2	r, r^3	s, r^2s	rs, r^3s
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
E	2	-2	0	0	0

3 The D_4 isotypic decomposition

The following theorem is stated in HCSM-05 [5]. We recall it here for self-containedness.

Theorem 3.1 (D_4 isotypic decomposition of V_{14}). *Under D_4 , the complexified 14-mode multiplet decomposes as*

$$V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (11)$$

with multiplicities $(m_{A_1}, m_{A_2}, m_{B_1}, m_{B_2}, m_E) = (3, 1, 2, 2, 3)$.

Proof. Direct decomposition via the character formula of HCSM-05. We record the character of the permutation module on the 14 modes:

$$\chi_{V_{14}} = (14, 2, 0, 2, 2) \quad (12)$$

on conjugacy classes $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$. Then

$$m_\rho = \frac{1}{8} \sum_C |C| \chi_{V_{14}}(C) \chi_\rho(C)^*, \quad (13)$$

giving $(3, 1, 2, 2, 3)$. Dimension check: $3 + 1 + 2 + 2 + 6 = 14$. $\square$

Remark 3.2 (Isotypic decomposition). The isotypic decomposition of V_{14} is

$$V_{14} = \bigoplus_{\rho} V_{\rho} \otimes M_{\rho}, \quad (14)$$

where V_{ρ} is the irreducible representation space of ρ and M_{ρ} is the multiplicity space of dimension m_{ρ} . Explicitly:

- $V_{A_1} \otimes M_{A_1}$: $\dim V_{A_1} = 1, \dim M_{A_1} = 3$.

- $V_{A_2} \otimes M_{A_2}$: $\dim V_{A_2} = 1$, $\dim M_{A_2} = 1$.
- $V_{B_1} \otimes M_{B_1}$: $\dim V_{B_1} = 1$, $\dim M_{B_1} = 2$.
- $V_{B_2} \otimes M_{B_2}$: $\dim V_{B_2} = 1$, $\dim M_{B_2} = 2$.
- $V_E \otimes M_E$: $\dim V_E = 2$, $\dim M_E = 3$.

The multiplicity space M_{A_1} of dimension 3 is the natural home of the color algebra; the multiplicity space M_{B_1} of dimension 2 is the natural home of the weak-isospin algebra.

4 The D_4 commutant

4.1 Definition and Schur's lemma

Definition 4.1 (D_4 commutant). The D_4 commutant on V_{14} is the algebra of D_4 -equivariant endomorphisms,

$$\text{Comm}(D_4, V_{14}) = \{X \in \text{End}_{\mathbb{C}}(V_{14}) : [X, R] = [X, S] = 0\}. \quad (15)$$

Theorem 4.2 (Schur decomposition of the commutant). *For a D_4 -representation $V = \bigoplus_{\rho} V_{\rho} \otimes M_{\rho}$ with $\dim M_{\rho} = m_{\rho}$, the commutant is*

$$\text{Comm}(D_4, V) \cong \bigoplus_{\rho} \text{End}_{\mathbb{C}}(M_{\rho}) = \bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C}). \quad (16)$$

Proof. Standard Schur's lemma. Since the V_{ρ} are pairwise non-isomorphic, every D_4 -equivariant endomorphism preserves the isotypic decomposition; on each block $V_{\rho} \otimes M_{\rho}$, an equivariant endomorphism must act as $\text{id}_{V_{\rho}} \otimes A_{\rho}$ with $A_{\rho} \in \text{End}(M_{\rho})$. The general element is thus $\bigoplus_{\rho} \text{id}_{V_{\rho}} \otimes A_{\rho}$. $\square$

Theorem 4.3 (Commutant dimension). *The D_4 commutant on the complexified 14-mode multiplet has complex dimension*

$$\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = \sum_{\rho} m_{\rho}^2 = 3^2 + 1^2 + 2^2 + 2^2 + 3^2 = 27. \quad (17)$$

Proof. Immediate from Theorem 4.2 and the multiplicities $(3, 1, 2, 2, 3)$. $\square$

4.2 The endomorphism algebra

Theorem 4.4 (Endomorphism algebra structure). *The commutant decomposes explicitly as*

$$\text{Comm}(D_4, V_{14}) \cong M_3(\mathbb{C}) \oplus M_1(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C}), \quad (18)$$

with $\mathbb{C}$ -dimensions $9 + 1 + 4 + 4 + 9 = 27$.

Proof. Immediate from Theorems 4.2 and 4.3. The five blocks correspond to the five irreps A_1, A_2, B_1, B_2, E in that order. $\square$

Remark 4.5 (The block decomposition and the physical sectors). The five matrix blocks are the natural Lie-algebraic factors:

- $M_3(\mathbb{C})$ on the A_1 multiplicity space: complex dimension 9, real dimension 18.

- $M_1(\mathbb{C})$ on the A_2 multiplicity space: complex dimension 1, real dimension 2.
- $M_2(\mathbb{C})$ on the B_1 multiplicity space: complex dimension 4, real dimension 8.
- $M_2(\mathbb{C})$ on the B_2 multiplicity space: complex dimension 4, real dimension 8.
- $M_3(\mathbb{C})$ on the E multiplicity space: complex dimension 9, real dimension 18.

The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ embeds in this algebra, with $\mathfrak{su}(3)$ in the A_1 block, $\mathfrak{su}(2)$ in the B_1 block, and $\mathfrak{u}(1)^5$ spanned by the identity elements of all five blocks.

5 The gauge algebra

5.1 The $\mathfrak{su}(3)$ subalgebra on the A_1 multiplicity space

The A_1 multiplicity space $M_{A_1} \cong \mathbb{C}^3$ is the home of the color algebra. Let $\{u_1, u_2, u_3\}$ be an orthonormal basis of M_{A_1} .

Definition 5.1 (Embedding map). The embedding of the A_1 block endomorphism algebra is

$$\phi_{A_1} : M_3(\mathbb{C}) \rightarrow \text{End}_{\mathbb{C}}(V_{14}), \quad \phi_{A_1}(X) = \sum_{a,b=1}^3 u_a X_{ab} u_b^\dagger, \quad (19)$$

extended by zero on the other isotypic blocks.

Proposition 5.2 (Homomorphism). ϕ_{A_1} is a unital algebra homomorphism and a Lie algebra homomorphism:

$$\phi_{A_1}(XY) = \phi_{A_1}(X)\phi_{A_1}(Y), \quad \phi_{A_1}([X, Y]) = [\phi_{A_1}(X), \phi_{A_1}(Y)]. \quad (20)$$

Proof. Direct computation. See also [8] for the analogous construction on the real mode space. $\square$

Theorem 5.3 ($\mathfrak{su}(3)$ subalgebra). The traceless anti-Hermitian subalgebra of $M_3(\mathbb{C})$,

$$\mathfrak{su}(3) = \{X \in M_3(\mathbb{C}) : X^\dagger = -X, \text{tr}(X) = 0\}, \quad (21)$$

embeds in $\text{Comm}(D_4, V_{14})$ via ϕ_{A_1} as a real 8-dimensional compact Lie subalgebra.

Proof. The image $\phi_{A_1}(\mathfrak{su}(3))$ is contained in the A_1 block of the commutant, and ϕ_{A_1} preserves brackets by the proposition above. The Lie algebra $\mathfrak{su}(3)$ is generated by the anti-Hermitian traceless 3×3 matrices, of real dimension 8. Its standard physics form uses the Gell-Mann matrices λ^a ($a = 1, \dots, 8$) with Hermitian generators $T^a = \lambda^a/2$ satisfying

$$[T^a, T^b] = i \sum_c f^{abc} T^c, \quad (22)$$

where f^{abc} are the $\mathfrak{su}(3)$ structure constants. The embedding in the commutant is

$$T_c^a = \phi_{A_1}(\lambda^a/2), \quad a = 1, \dots, 8. \quad (23)$$

$\square$

Remark 5.4 (Physical color as input). The identification of this $\mathfrak{su}(3)$ factor with the physical color algebra $\mathfrak{su}(3)_c$ is a physical input of the framework, not a derivation. The framework derives that a copy of $\mathfrak{su}(3)$ acts naturally on the A_1 multiplicity space; the identification with color is the same identification the Standard Model makes.

5.2 The $\mathfrak{su}(2)$ subalgebra on the B_1 multiplicity space

The B_1 multiplicity space $M_{B_1} \cong \mathbb{C}^2$ is the home of the weak isospin algebra. Let $\{v_1, v_2\}$ be an orthonormal basis.

Definition 5.5 (Embedding map). $\phi_{B_1} : M_2(\mathbb{C}) \rightarrow \text{End}_{\mathbb{C}}(V_{14})$ is defined by

$$\phi_{B_1}(X) = \sum_{a,b=1}^2 v_a X_{ab} v_b^\dagger, \quad (24)$$

extended by zero on the other isotypic blocks.

Theorem 5.6 ($\mathfrak{su}(2)$ subalgebra). *The traceless anti-Hermitian subalgebra $\mathfrak{su}(2) \subset M_2(\mathbb{C})$ embeds in $\text{Comm}(D_4, V_{14})$ via ϕ_{B_1} as a real 3-dimensional compact Lie subalgebra.*

Proof. Same structure as Theorem 5.3. The Lie algebra $\mathfrak{su}(2)$ is generated by the anti-Hermitian traceless 2×2 matrices; the standard physics basis uses the Pauli matrices σ^i , with Hermitian generators $T^i = \sigma^i/2$ satisfying $[T^i, T^j] = i\epsilon^{ijk}T^k$. $\square$

Remark 5.7 (Physical weak isospin as input). The identification of this $\mathfrak{su}(2)$ factor with the physical weak isospin algebra $\mathfrak{su}(2)_L$ is a physical input, not a derivation.

5.3 The $\mathfrak{u}(1)^5$ subalgebra from the isotypic projectors

Definition 5.8 (Isotypic projectors). For each irrep ρ of D_4 , the isotypic projector is

$$P_\rho = \frac{\dim \rho}{|D_4|} \sum_{g \in D_4} \chi_\rho(g)^* \rho_{14}(g). \quad (25)$$

The five projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$ satisfy $P_\rho P_{\rho'} = \delta_{\rho\rho'} P_\rho$ and $\sum_\rho P_\rho = \mathbb{I}_{14}$.

Theorem 5.9 ($\mathfrak{u}(1)^5$ subalgebra). *The five isotypic projectors span an abelian Lie subalgebra of $\text{Comm}(D_4, V_{14})$ isomorphic to $\mathfrak{u}(1)^5$, of real dimension 5.*

Proof. Each P_ρ commutes with the D_4 action by construction, so $P_\rho \in \text{Comm}(D_4, V_{14})$. The projectors are mutually orthogonal ($P_\rho P_{\rho'} = 0$ for $\rho \neq \rho'$) and sum to $\mathbb{I}_{14}$, giving a 5-dimensional span. As a Lie algebra with the commutator bracket, this span is abelian (all brackets vanish), hence isomorphic to $\mathfrak{u}(1)^5$. $\square$

5.4 Closure of the gauge algebra

Theorem 5.10 (Closure). *The image $\phi_{A_1}(\mathfrak{su}(3)) \oplus \phi_{B_1}(\mathfrak{su}(2)) \oplus \text{span}\{P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$ is a Lie subalgebra of $\text{Comm}(D_4, V_{14})$ isomorphic to*

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5, \quad (26)$$

of real dimension $8 + 3 + 5 = 16$.

Proof. We verify the brackets.

(1) $\mathfrak{su}(3)$ - $\mathfrak{su}(3)$ brackets. ϕ_{A_1} is a Lie algebra homomorphism, so $[\phi_{A_1}(X), \phi_{A_1}(Y)] = \phi_{A_1}([X, Y]) \in \phi_{A_1}(\mathfrak{su}(3))$.

(2) $\mathfrak{su}(2)$ - $\mathfrak{su}(2)$ brackets. Same as (1).

(3) $\mathfrak{su}(3)$ - $\mathfrak{su}(2)$ brackets. $\phi_{A_1}(\mathfrak{su}(3))$ is supported only on the A_1 multiplicity space; $\phi_{B_1}(\mathfrak{su}(2))$ is supported only on the B_1 multiplicity space. The two are orthogonal in $\text{End}_{\mathbb{C}}(V_{14})$, so $[\phi_{A_1}(X), \phi_{B_1}(Y)] = 0$.

(4) $\mathfrak{su}(3)$ - $\mathfrak{u}(1)^5$ brackets. Each projector P_ρ acts as the identity on the ρ -block and zero elsewhere. On the A_1 block, P_{A_1} acts as $\mathbb{I}_{M_{A_1}}$, which commutes with every element of $M_{A_1}(\mathbb{C})$ (in particular with $\phi_{A_1}(\mathfrak{su}(3))$). Similarly for $P_{A_2}, P_{B_1}, P_{B_2}, P_E$ on the A_1 block (they act as zero). Hence $[\phi_{A_1}(X), P_\rho] = 0$ for all $X \in \mathfrak{su}(3)$ and all ρ .

(5) $\mathfrak{su}(2)$ - $\mathfrak{u}(1)^5$ brackets. Same as (4).

(6) $\mathfrak{u}(1)^5$ - $\mathfrak{u}(1)^5$ brackets. The projectors are mutually commuting, so their span is abelian.

The algebra thus closes to $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$, a direct sum of the three factors, of real dimension $8 + 3 + 5 = 16$. $\square$

Remark 5.11 (Sub-algebra of the commutant). The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ is a 16-dimensional real Lie subalgebra of the 27-dimensional complex commutant $\bigoplus_\rho M_{m_\rho}(\mathbb{C})$. The commutant has real dimension 54 (as a real vector space), so the gauge algebra is a proper subspace; the remaining directions correspond to the “extra” structure not used by the gauge algebra (specifically, the traceless parts of the A_2 , B_2 , and E multiplicity blocks, which are also matrix algebras but not part of the framework’s gauge algebra).

6 Sector-coarse correspondence

The derivation above was on the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$. We now extend to the full 56-dimensional Hodge sector at $\lambda = 4$, establishing the sector-coarse correspondence.

6.1 The full Hodge sector

The full 56-dimensional sector at $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (27)$$

with D_4 decomposition (from HCSM-02 [3])

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (28)$$

Dimension check: $9 + 5 + 7 + 7 + 28 = 56$.

6.2 Sector operators

Definition 6.1 (Sector operators). On V_{56} , define the three sector operators

$$\mathbb{I} = P_{\Omega^0} + P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{F} = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad \mathcal{O} = P_{\Omega^0} - P_{\Omega^2}. \quad (29)$$

Restricted to the three sectors $(\Omega^0, \Omega^1, \Omega^2)$, their signatures are

$$\mathbb{I}|_{\text{sectors}} = (1, 1, 1), \quad \mathcal{F}|_{\text{sectors}} = (1, -1, 1), \quad \mathcal{O}|_{\text{sectors}} = (1, 0, -1). \quad (30)$$

6.3 The sector-coarse sub-commutant

Definition 6.2 (Sector-coarse sub-commutant). The sector-coarse sub-commutant is the space of operators $X \in \text{End}_{\mathbb{C}}(V_{56})$ satisfying:

- (i) $[X, R] = [X, S] = 0$ (D_4 -invariance);
- (ii) $X = P_{\Omega^0} X P_{\Omega^0} + P_{\Omega^1} X P_{\Omega^1} + P_{\Omega^2} X P_{\Omega^2}$ (sector block-diagonality);
- (iii) X acts as a scalar on each (ρ, Ω^k) block, where $\rho \in \{A_1, A_2, B_1, B_2, E\}$ and $k \in \{0, 1, 2\}$.

Theorem 6.3 (Sector-coarse correspondence). *The sector-coarse sub-commutant has complex dimension exactly $15 = 5 \times 3$ (five irreps $\times$ three Hodge sectors). It is spanned by the three sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ acting on each isotypic block.*

Proof. By Schur's lemma applied to each (ρ, Ω^k) block, a D_4 -invariant, sector-block-diagonal, multiplicity-scalar operator is determined by 15 complex scalar coefficients—one for each of the $5 \times 3 = 15$ blocks. Hence the space has complex dimension 15.

To see that the span of $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each block has the same dimension: for each fixed ρ , the restrictions of $(\mathbb{I}, \mathcal{F}, \mathcal{O})$ to the three sectors give three vectors in $\mathbb{C}^3$:

$$\mathbb{I}^{(\rho)} = (1, 1, 1), \quad \mathcal{F}^{(\rho)} = (1, -1, 1), \quad \mathcal{O}^{(\rho)} = (1, 0, -1). \quad (31)$$

The matrix with rows $(\mathbb{I}^{(\rho)}, \mathcal{F}^{(\rho)}, \mathcal{O}^{(\rho)})$ is

$$M = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}, \quad \det M = 1(1 - 0) - 1(-1 - 1) + 1(0 + 1) = 1 + 2 + 1 = 4 \neq 0. \quad (32)$$

So the three operators are linearly independent and span $\mathbb{C}^3$ on each ρ -block. Over the five irreps, the span has complex dimension $5 \times 3 = 15$. The two spaces coincide. $\square$

Corollary 6.4 (Orientation- $\mathbb{Z}_2$). *A sector-coarse functional Y on V_{56} is a complex linear combination*

$$Y = \sum_{\rho} (\alpha_{\rho} \mathbb{I} + \beta_{\rho} \mathcal{F} + \gamma_{\rho} \mathcal{O}) P_{\rho}, \quad (33)$$

of the three sector operators on each isotypic block. Under the sign flip $\mathcal{O} \mapsto -\mathcal{O}$, the multiset of physical sector values is preserved, up to relabeling of Ω^0 and Ω^2 .

Proof. Immediate from Theorem 6.3 and the antisymmetry of $\mathcal{O}$ under the D_4 orientation flip $\Omega^0 \leftrightarrow \Omega^2$. The full orientation- $\mathbb{Z}_2$ theorem is the subject of HCSM-09 [7]. $\square$

7 Inputs and derived

7.1 Derived

The following results are derived from the substrate and the D_4 action alone:

1. The D_4 isotypic decomposition $V_{14} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$ (Theorem 3.1; from HCSM-05).
2. The complex dimension of the D_4 commutant, $\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = 27$ (Theorem 4.3).

3. The endomorphism algebra decomposition $\text{Comm}(D_4, V_{14}) \cong M_3(\mathbb{C}) \oplus M_1(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus M_3(\mathbb{C})$ (Theorem 4.4).
4. The embedding of $\mathfrak{su}(3)$ in the A_1 block (Theorem 5.3).
5. The embedding of $\mathfrak{su}(2)$ in the B_1 block (Theorem 5.6).
6. The embedding of $\mathfrak{u}(1)^5$ from the isotypic projectors (Theorem 5.9).
7. The closure of the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (Theorem 5.10).
8. The sector-coarse correspondence on the 56-dimensional Hodge sector: 15-dimensional subcommutant (Theorem 6.3).

7.2 Inputs

The following are physical inputs, not derivations:

1. **Input 1 (physical identification).** The identification of the A_1 factor with the physical color algebra $\mathfrak{su}(3)_c$ and the B_1 factor with the physical weak isospin algebra $\mathfrak{su}(2)_L$. The framework derives that an $\mathfrak{su}(3)$ factor and an $\mathfrak{su}(2)$ factor act naturally on the multiplicities; the identification of these factors with the physical gauge groups is the same identification the Standard Model makes.
2. **Input 2 (physical hypercharge).** The physical hypercharge Y is a specific linear combination of the three sector operators $(\mathbb{I}, \mathcal{F}, \mathcal{O})$ on the A_1 and B_1 blocks. The specific coefficients are determined by the Standard Model content and are the subject of HCSM-07 [6].

7.3 What is not claimed

We do not claim to derive the Standard Model gauge group $\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y$ from first principles. We derive the Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ as a subalgebra of the D_4 commutant. The reduction to the physical gauge algebra involves the physical inputs above.

The framework's claim is: *any* physically realized gauge theory with Standard Model content and the D_4 structure of the 14-mode multiplet must be compatible with a subalgebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ of the D_4 commutant. The physical content of the gauge algebra is then selected by Input 1 and Input 2.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSMcompanion repository at `12-VERIFICATION/HCSM-06_verification`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the 14-mode multiplet and its D_4 permutation representation, and verifies:

1. The D_4 isotypic decomposition of V_{14} (Theorem 3.1).
2. The commutant dimension $\dim_{\mathbb{C}} \text{Comm}(D_4, V_{14}) = 27$ (Theorem 4.3).
3. The endomorphism algebra decomposition $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$ (Theorem 4.4).
4. The $\mathfrak{su}(3)$ embedding in the A_1 block (Theorem 5.3).

5. The $\mathfrak{su}(2)$ embedding in the B_1 block (Theorem 5.6).
6. The $\mathfrak{u}(1)^5$ subalgebra from the isotypic projectors (Theorem 5.9).
7. The bracket closure of $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (Theorem 5.10).
8. The 15-dimensional sector-coarse sub-commutant (Theorem 6.3).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSMpaper stack.

9 Conclusion

This paper has established the *sector-coarse correspondence* of the Hodge Complex Standard Model: the D_4 isotypic sectors of the 14-mode multiplet at $\lambda = 4$ correspond to the gauge algebra structure through the endomorphism algebra of their multiplicity spaces.

The 14-mode multiplet decomposes under D_4 as $3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E$. The D_4 commutant on the complexified multiplet is the endomorphism algebra $\bigoplus_{\rho} M_{m_{\rho}}(\mathbb{C})$, of complex dimension 27. The gauge algebra

$$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5 \tag{34}$$

embeds in the commutant: $\mathfrak{su}(3)$ acts on the A_1 multiplicity space (dim 3), $\mathfrak{su}(2)$ on the B_1 multiplicity space (dim 2), and the five isotypic projectors span the $\mathfrak{u}(1)^5$ center.

The derivation is HCSM-native: it uses only the D_4 action on the 14-mode multiplet at $\lambda = 4$ and the standard endomorphism algebra structure. No imported machinery (no path integrals, no SYK kernels, no gauge fields as fundamental objects) is used.

On the full 56-dimensional Hodge sector at $\lambda = 4$, the sector-coarse correspondence is that the space of D_4 -invariant, sector-block-diagonal, multiplicity-scalar operators is exactly 15-dimensional, spanned by the sector operators $\{\mathbb{I}, \mathcal{F}, \mathcal{O}\}$ on each of the five isotypic blocks. This 15-dimensional space is the natural home of the physical hypercharge functional, whose specific coefficients are determined by the Standard Model content and are the subject of HCSM-07.

The identification of the derived $\mathfrak{su}(3)$ and $\mathfrak{su}(2)$ factors with the physical color and weak-isospin gauge groups is a physical input, the same input the Standard Model itself makes. The framework derives the algebraic structure that any physically realized gauge theory with the Standard Model content and the D_4 structure of the 14-mode multiplet must be compatible with.

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- [9] I. Schur, Neue Begründung der Theorie der Gruppencharaktere, Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (1905), 406–432.

A Web Crawler Metadata

A.1 Dublin Core

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DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
DC.subject: Hodge Complex Standard Model; HCSM; D4 commutant; endomorphism
  algebra; Schur's lemma; gauge algebra; su(3); su(2); u(1)^5;
  sector-coarse correspondence; 14-mode multiplet; 8x8 torus
DC.description: The Hodge Complex Standard Model assigns the 14 modes at
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  commutant on the complexified 14-mode multiplet, through the
  endomorphism algebra of the D4 isotypic multiplicity spaces. The D4
  commutant is 27-dimensional complex;  $\text{su}(3)$  embeds in the A1 block,
   $\text{su}(2)$  in the B1 block, and  $\text{u}(1)^5$  is spanned by the five isotypic
  projectors. On the full 56-dimensional Hodge sector at  $\lambda = 4$ , the
  sector-coarse correspondence is that the space of D4-invariant,
  sector-block-diagonal, multiplicity-scalar operators is exactly
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DC.identifier: HCSM-06-2026-09

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A.2 Schema.org JSON-LD

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    "name": "Entropia Research Institute"
  }
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"inLanguage": "en",
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  "D4 commutant",
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],
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"identifier": "HCSM-06-2026-09",
"isPartOf": {
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  "name": "HCSM White Paper Series"
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A.3 BibTeX for Automated Indexing

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  title = {{HCSM-06}: Sector-Coarse Correspondence},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, D4 commutant,
    endomorphism algebra, Schur's lemma, gauge algebra, su(3), su(2),
    u(1)^5, sector-coarse correspondence, 14-mode multiplet},
  abstract = {This paper derives the gauge algebra  $\mathfrak{su}(3) + \mathfrak{su}(2) + \mathfrak{u}(1)^5$  as a Lie subalgebra of the D4 commutant on the complexified
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14-mode multiplet at $\lambda = 4$. The D4 commutant is 27-dimensional complex; $\mathfrak{su}(3)$ embeds in the A1 block, $\mathfrak{su}(2)$ in the B1 block, and $\mathfrak{u}(1)^5$ is spanned by the five isotypic projectors. On the 56-dimensional Hodge sector, the sector-coarse correspondence is a 15-dimensional sub-commutant.}

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